

Dynamic Reconstruction and Assessment Model of Road Accident Risk Domain Incorporating Spatial Attenuation Effects and Multidimensional Coupling

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Abstract: Traditional identification of accident-prone road sections relies on fixed-unit statistics, which has disadvantages that include missing spatial continuity of risk, boundary distortion, and information “averaging,” and struggles to support differentiated safety management. To address this issue, this study proposes a road accident risk field reconstruction model integrating dynamic spatial attenuation and probability-severity dual-dimensional coupling. First, based on the information diffusion principle and Gaussian kernel density estimation (KDE), a framework for converting discrete accident points into a continuous risk probability field is constructed, and a dynamic standard deviation function is proposed. This function enables the Gaussian kernel standard deviation to dynamically adapt to road design speed and cross-sectional type (with/without median divider), clarifying its physical correlation with drivers’ sight distance requirements. Second, by integrating casualties and direct economic losses, a continuous accident severity field is established through an accident equivalent loss function; the two fields are discretized using the quantile method, and combined with a risk matrix to generate comprehensive risk ratings (Levels I–V) via coupling. Verification using accident data from a Class II mountain highway in Guangxi shows the following: (1) the boundaries of the risk field generated by the model fit the road alignment well, with a probability field gradient smoothness of 0.0068, which can mitigate the boundary effect and step effect of the fixed-unit method; (2) compared with KDE with fixed bandwidth (0.35 km) and fixed-length segmentation method (0.5 km), the area under the curve (AUC) of the model’s probability field reaches 0.9641 (increased by 12.79% and 15.31%, respectively), the AUC of the severity field is 0.8669 (increased by 15.24% and 5.88%, respectively), and the Pearson correlation coefficient between the probability field and accident data is 0.4772 (increased by 11.01% and 3.77%, respectively), indicating that the identification accuracy and data fit have been improved; and (3) dual-dimensional coupling can distinguish risk patterns of high-frequency low-loss, low-frequency high-loss, and high-frequency high-loss, providing a quantitative basis for differentiated management. This model promotes the evolution of road risk assessment from discrete statistics to continuous field analysis, and can offer technical support for the optimal allocation of safety resources. DOI: [10.1061/JTEPBS.TEENG-9585](https://doi.org/10.1061/JTEPBS.TEENG-9585). © 2026 American Society of Civil Engineers.

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Introduction

Road traffic accidents represent a severe global public safety and health challenge, causing numerous casualties and substantial socioeconomic losses that continue to pose a threat to sustainable development (World Health Organization 2024). Against this backdrop, accurately identifying high-risk road segments, diagnosing safety hazards, and optimizing the allocation of safety resources are core tasks for improving road safety management (Ziakopoulos and Yannis 2020). Traditional methods for identifying accident-prone locations (black spots) primarily rely on fixed-length discrete unit statistical approaches (e.g., 500-m road segments), which have some notable limitations: (1) they are highly sensitive to the unit division scale, which may lead to inconsistencies in risk assessment, resulting in spatial boundary effects and abrupt changes in the risk field (Thakali et al. 2015; Zheng et al. 2024); (2) they struggle to reflect the continuous impacts of road design elements such as horizontal and vertical alignments, disrupting the true influence scope of systemic risks (Ziakopoulos and Yannis 2020); and (3) they inadequately characterize the spatiotemporal lag effects of driving behavior, leading to distorted risk representation (Thakali et al. 2015).

To address these shortcomings, academic circles have focused on continuous risk field modeling in recent years, exploring methods including information diffusion, Gaussian kernel density

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estimation (KDE), Bayesian hierarchical models, and deep learning, while attempting to couple multidimensional characteristics such as probability and severity (Mullakkal-Babu et al. 2020; Clemente et al. 2024; Mogyorósi et al. 2024; Zuo et al. 2025). However, existing methods still have key deficiencies: Core parameters lack engineering and physical significance, robustness to small samples is insufficient, multidimensional risk coupling mechanisms are incomplete, and systematic comparisons with mature benchmark methods have not been conducted (Arun et al. 2021; Katicha and Flintsch 2023; Bonera et al. 2024). Through a systematic review of research progress and clarification of existing gaps, this study proposes a continuous risk field reconstruction and evaluation framework with clear theory, interpretable parameters, and adaptability to small-sample conditions.

Literature Review

Evolution and Challenges of Continuous Spatial Risk Modeling Methods

Gaussian KDE is a mainstream method for constructing continuous risk surfaces (Katicha and Flintsch 2023), but its bandwidth parameters are mostly purely data-driven and lack engineering interpretability (Thakali et al. 2015). Although some studies have improved KDE by introducing variable bandwidths (Zheng et al. 2024) or Bayesian frameworks (KDEB) (Katicha and Flintsch 2023), the fundamental disconnect between parameters and engineering practice has not been resolved. Two-stage frameworks combining clustering algorithms such as HDBSCAN (hierarchical density-based spatial clustering of applications with noise) with Geographic Information Systems (GIS) have improved the automation level and stability of high-risk road segment localization (Wang et al. 2023).

Bayesian statistical models can capture spatial heterogeneity through random effects and provide stable estimates under small-sample conditions (Aziz et al. 2024; Feng et al. 2025). Among them, the spatial Poisson log-normal (SPLN) model exhibits the best goodness of fit (Guo et al. 2018) and the full Bayesian random parameter multivariate tobit (RPMV-Tobit) model can handle the correlation between different collision types (Guo et al. 2018, 2024). While these methods are statistically rigorous, they feature complex structures, high computational costs, and relatively high barriers to engineering application.

With the advancement of computing power, machine learning and deep learning methods have demonstrated considerable potential (Hemalatha 2025). For example, random forests and support vector machines have been used for real-time accident risk prediction (Yang et al. 2023), and conditional generative adversarial networks (CGANs) have been employed to predict spatiotemporal impact heatmaps of traffic accidents (Wen et al. 2022). However, most of these models are “black boxes” with poor interpretability and rely heavily on large-scale high-quality data. Their stability and generalization ability are limited in scenarios with scarce accident data. Therefore, how to construct models that possess both predictive power and interpretability under data-limited conditions remains a key issue.

Deepening Understanding of Spatiotemporal Heterogeneity and Integration of Multidimensional Risk Assessment

Existing studies have confirmed that road safety risks exhibit significant spatiotemporal heterogeneity, such as the temporal heterogeneity of accident delays (Wang et al. 2025) and the seasonal fluctuations of accident risks for vulnerable road users (Abdulrazaq and Fan 2026). These studies have advanced the understanding of the time-varying laws of risks.

However, most models focus on single-dimensional accident frequency prediction and fail to systematically couple the “probability-severity” dual dimensions—a comprehensive risk assessment should integrate both. Some studies have attempted to construct comprehensive risk indicators (Mullakkal-Babu et al. 2020; Jia et al. 2025) or use surrogate safety measures (SSMs) to characterize risks (Yang et al. 2021), but the logical framework for the seamless integration of severity information with continuous risk fields has not been clarified, nor has weight sensitivity analysis for coupling been conducted.

Research Gaps and Objectives

A review of existing studies identifies three core issues in continuous risk field modeling: insufficient parameter interpretability, lack of small-sample adaptability, and incomplete multidimensional risk coupling mechanisms. To address these, this study proposes a new framework integrating information diffusion theory and physically meaningful guided Gaussian KDE, with core innovations and contributions reflected in three aspects.

First, the theoretical mechanism and parameter interpretability are deepened. Existing models [e.g., Bayesian hierarchical models (Aziz et al. 2024; Feng et al. 2025), STAA framework (De Angeli et al. 2022)] focus on macro statistical fitting, with parameters lacking clear engineering and physical significance. This study couples information diffusion theory with a Gaussian spatial attenuation model to clarify the continuous diffusion logic of risk point sources. An innovative dynamic standard deviation function $\sigma(v)$ is proposed, establishing an explicit correlation with road design speed and alignment conditions based on sight distance models. This enables the core parameter σ to directly reflect the spatial requirements of driving safety, realizing a shift from “data-driven/empirical” to “mechanism-guided/engineering-oriented” and enhancing the model’s interpretability and engineering applicability.

Second, small-sample adaptability and robustness are enhanced. Traditional fixed-bandwidth KDE and complex data-driven models tend to exhibit unstable performance in scenarios with sparse accident data. This framework leverages information diffusion theory (Chongfu 1997) to handle incomplete information, and the dynamic parameter $\sigma(v)$ is determined based on road physical attributes. This reduces reliance on large volumes of historical accident data, providing a robust risk assessment solution for data-scarce scenarios such as mountainous highways.

Third, a multidimensional risk assessment system is constructed. Existing methods mostly rely on single-dimensional accident frequency and struggle to integrate the dual attributes of probability-consequence. This study establishes a dual-mode risk field of probability-severity, enabling independent continuous spatial characterization of the two dimensions. Combined with loss conversion coefficients related to accident compensation standards and a decision-oriented coupling model prioritizing severity, key low-frequency high-loss road sections are systematically identified. Complemented by weight sensitivity analysis, the evaluation results are more aligned with the core needs of safety management.

The structure of this paper is arranged as follows: We first elaborate on the theoretical basis and construction process of the model. Next, the case data are introduced as well as preprocessing procedures and experimental design. We present the model results, sensitivity analysis, and comparative verification, and conclude with a discussion of the model’s limitations and future research directions.

Theoretical Basis and Model Construction

Core Limitations of Traditional Discretization Methods

The fixed-length unit statistical method counts accident frequencies by artificially dividing discrete units and has three inherent limitations. First, it exhibits significant boundary effects, truncating the continuous impact of risks and leading to abrupt risk changes at unit boundaries. Second, it suffers from distorted information gradients, obscuring the fine-grained risk differences within units. Third, it disrupts systemic risk factors, overlooking the continuous influence range of road geometric defects. Fig. 1 illustrates that differences in unit scales may lead to shifts in risk ratings, highlighting the scale sensitivity and representation distortion issues of traditional methods.

Theoretical Framework of Risk Field Reconstruction

Theoretical Connection between Information Diffusion and Gaussian Spatial Attenuation

To address the problem of sparse small-sample data and achieve continuous spatial characterization of risks, this study couples the principle of information diffusion with Gaussian kernel estimation. The core of information diffusion theory (Chongfu 1997) is to regard a single accident point as incomplete sampling of a continuous risk distribution. Through a diffusion function, information from discrete points is allocated to the surrounding area, supplementing missing risk information in small samples and realizing the transformation from discrete samples to continuous risk distribution. The Gaussian spatial attenuation effect is consistent with the propagation characteristics of road risks, where physical impacts decrease smoothly with distance, serving as a “physical carrier” for diffusion. These two approaches complement each other: Information diffusion provides a framework for handling small-sample uncertainty, solving the problem of discrete-to-continuous transformation; Gaussian kernel endows diffusion with a mathematical form of explicit attenuation rate, addressing the problem of “reasonable information allocation” and ensuring the theoretical rigor and physical rationality of continuous risk characterization under small-sample conditions.

From the perspective of a single accident point, the mathematical expression is as follows: Let the i th accident point be located at the linear coordinate x_i on the road; its risk intensity contribution $P_{i(x)}$ to any point x on the road is defined as a Gaussian function

$$P_{i(x)} = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-x_i)^2}{2\sigma^2}\right) \quad (1)$$

where x_i = linear coordinate of the i th accident point in meters; x = linear coordinate of any point to be evaluated on the road in meters; and σ = standard deviation of the Gaussian kernel function in meters, where σ serves as a key parameter controlling the risk attenuation rate. A larger σ value indicates a wider risk impact range and gentler attenuation, while the impact becomes more concentrated with steeper attenuation when σ is smaller.

As shown in Fig. 2, the Gaussian superposition curve of accident points indicates that risk intensity attenuates continuously with distance, while traditional discrete statistical methods treat it as a piecewise constant function. This simplification ignores the spatial radiation effect of hazard sources and obscures the risk superposition mechanism between accident points. For example, the overlapping area of risk domains of adjacent accidents may be underestimated as a low-risk area.

Construction of Dynamic Standard Deviation Function

The core difference between this model and traditional fixed-bandwidth KDE lies in the determination mechanism of parameter σ . This study proposes a dynamic standard deviation function $\sigma(v, T)$, whose dynamic nature is reflected in that the σ value is jointly determined by road design speed v and cross-sectional type T (with/without median divider), rather than changing with real-time vehicle speed. Design speed has high stability and significant engineering application value, which avoids the defect of strong spatiotemporal volatility of real-time vehicle speed and meets the core goal of engineering operability of the model.

The logical basis for function construction lies in the correlation between the spatial impact range of a hazard source and the driver's sight distance, which refers to the minimum spatial distance required for hazard identification and safe operation. Based on the risk perception-response theory of driving behavior (Ba et al. 2016), drivers have incompressible spatiotemporal response requirements, such that the impact range of hazard sources must match this spatial requirement and is subject to dual boundary constraints. One is the lower bound constraint, requiring the impact range to not be less than the sight distance—otherwise, incomplete coverage of risk hazards may lead to assessment underestimation. The other is the upper bound constraint, dictating the impact range to not exceed the sight distance; drivers cannot perceive risks beyond the sight distance, so the actual effect of hazard sources can be ignored. The critical characteristic of the impact boundary of hazard sources is the driver's response capacity threshold, which shares a

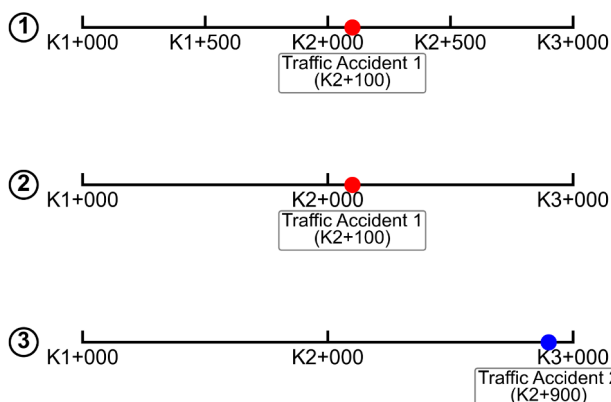


Fig. 1. Mechanism of risk field distortion caused by road section division.

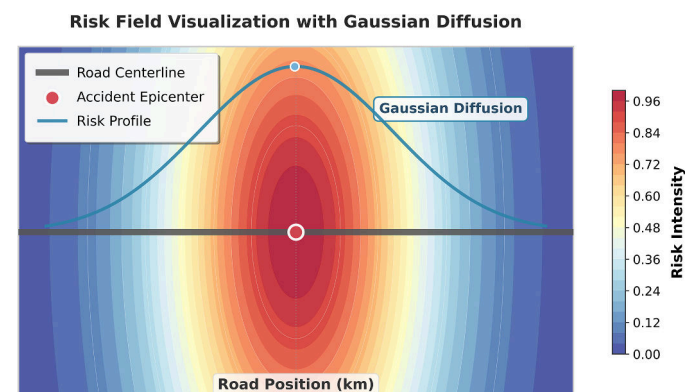


Fig. 2. Schematic representation of risk degree diminution and causal distribution spectrum Gaussian model focused on incident location.

natural unity in physical meaning with the engineering threshold of sight distance.

The statistical 3σ principle (Pukelsheim 1994) is introduced to establish the relationship $D = 3\sigma$ (i.e., $\sigma = D/3$), realizing the correspondence between the engineering impact range and statistical attenuation parameters. This assumption is an engineering simplification, and the model's robustness will be verified later through sensitivity analysis (testing values of 2σ and 4σ).

The expression of the dynamic standard deviation function is

$$\sigma(v, T) = \begin{cases} \frac{D_{SSD}(v)}{3}, & T = \text{with median divider} \\ \frac{D_{PSD}(v)}{3}, & T = \text{without median divider} \end{cases} \quad (2)$$

where $D_{SSD}(v)$ = stopping sight distance; and $D_{PSD}(v)$ = passing sight distance, both in meters.

Determination of Dual-Mode Risk Impact Radius

Mode 1 (Road with Median Divider). Mode 1 is applicable to expressways, Class I highways, etc. These roads feature physical isolation between opposite traffic flows, with the dominant risk being conflicts among same-direction traffic. The risk impact radius D_1 is calculated using the stopping sight distance formula specified in the Code for Safety Evaluation of Highway Projects (JTG B05-2015) [Eq. (3)]. Adverse working conditions such as wet road surfaces ($f = 0.29$) and downhill slopes ($i = -3\%$) are selected to ensure a safety margin

$$D_1 = D_{SSD} = \frac{v^2}{254(f \pm i)} + \frac{t_r \cdot v}{3.6} \quad (3)$$

where v = vehicle speed (km/h); f = longitudinal road friction coefficient, with a typical value of 0.29 for wet road surfaces; i = longitudinal road slope in percentage (%), where positive values correspond to uphill slopes and negative values to downhill slopes; and t_r refers to reaction time (s) from hazard detection to braking action, with a recommended value of 2.5 s according to the code.

The corresponding dynamic standard deviation is $\sigma_1 = D_1/3$, and typical values are shown in Table 1.

Mode 2 (Road without Median Divider). Mode 2 is applicable to Class II and lower highways, where opposite-direction collision risks exist, and overtaking is a typical high-risk scenario. The risk impact radius D_2 is determined by the PSD, which refers to the minimum distance required for a vehicle to safely complete the entire overtaking process. Its calculation formula and parameter values strictly follow the relevant provisions of the Code for Highway Alignment Design (JTG D20-2017)

$$D_2 = D_{PSD} = \left(v_2 t_1 + \frac{1}{2} a t_1^2\right) + v_1 t_2 + v_1 t_3 + v_3(t_1 + t_2 + t_3) \quad (4)$$

Table 1. Risk impact radius D_1 for roads with median divider (based on stopping sight distance)

Design speed (km/h)	Stopping sight distance (m)	Dynamic standard deviation σ_1 (m)
120	210	70.0
100	160	53.3
80	110	36.7
60	75	25.0

Note: Calculated under the condition of $f = 0.29$ and $i = -3\%$ (adverse downhill condition).

where v_2 = speed of the overtaken vehicle (m/s); t_1 = acceleration time (s), which is the time from the start of acceleration to reaching the overtaking speed; a = acceleration (m/s^2), indicating the rate of speed change during acceleration; v_1 = speed of the overtaking vehicle (m/s); t_2 = the overtaking time (s), which is the time required from the start to the completion of overtaking; t_3 = safety distance time (s), which is the time required to maintain a safe distance after overtaking; and v_3 = speed of oncoming vehicles (m/s).

In this study, the risk impact radius D_2 is taken as the "general value" specified in the code, with the corresponding dynamic standard deviation given by $\sigma_2 = D_2/3$. The specific values are presented in Table 2.

Through the aforementioned construction, the Gaussian kernel parameter σ is endowed with clear traffic engineering significance, thereby achieving in-depth integration of the mathematical model with road physical attributes and design standards and laying the foundation for developing a continuous risk field with engineering interpretability.

Dual-Dimensional Risk Field Coupling Model

Construction of Risk Probability Intensity Field

The risk contribution function of a single accident is given by Eq. (1). After normalization, the total risk probability intensity is the superposition of Gaussian distributions for all accident points

$$P(x) = \sum_{i=1}^N P_{i(x)} \quad (5)$$

The risk probability field generated by superposition can naturally characterize core risk areas (formed by the dense superposition of accident points), risk gradient attenuation zones, and isolated secondary risk peaks, thereby avoiding the "averaging" limitation of fixed-unit statistics.

Construction of Accident Severity Field

To comprehensively reflect the consequences of casualties and economic losses caused by accidents, a standardized accident equivalent loss L_e is constructed. For each accident i , the equivalent loss is calculated as follows:

To unify the dimensions of casualties and economic losses, an equivalent loss function is established as follows:

$$L_{e,i} = (\alpha \cdot I_{s,i} + \beta \cdot I_{f,i}) + \gamma \cdot D_{m,i} \quad (6)$$

where $I_{s,i}$ = number of injured persons obtained from accident records; $I_{f,i}$ = number of fatalities obtained from accident records; $D_{m,i}$ = direct economic loss in ¥10,000; and α and β are casualty loss conversion coefficients that convert the consequences of

Table 2. Risk impact radius D_2 for roads without median divider (based on passing sight distance)

Design speed (km/h)	Overtaking sight distance minimum value (m)		Dynamic standard deviation σ_2 (m)
	General value	Limit value	
80	550	350	183.3
60	350	250	116.7
40	200	150	66.7
30	150	100	50.2
20	100	70	33.3

Note: DPSD is taken from the "general value" in the code.

Table 3. Risk scoring table for severity-probability level combinations

Severity/probability	Low (L = 0)	Medium (L = 1)	High (L = 2)
High (S = 2)	7	8	9
Medium (S = 1)	4	5	6
Low (S = 0)	1	2	3

casualties of different severities into comparable economic equivalents, whose values can be calibrated based on the judicial practice of personal injury compensation for local road traffic accidents, local standards, or typical cases. This proportional relationship essentially reflects the general social recognition of the differentiated socioeconomic costs of casualty accidents, and this assignment has a certain degree of subjectivity and regionality, and the sensitivity of its values to the model results will be discussed in detail later. Also in Eq. (6), γ = economic loss weight (set to $\gamma = 1$ to unify the loss dimension); and $L_{e,i}$ = standardized accident severity indicator integrating multidimensional consequences.

Continuing the parameter system of the probability field (to ensure consistent spatial attenuation characteristics), the accident severity intensity field $I_{\text{sev}(x)}$ is constructed as follows:

$$I_{\text{sev}(x)} = \sum_{i=1}^N L_{e,i} \cdot \exp\left(-\frac{(x-x_i)^2}{2\sigma^2}\right) \quad (7)$$

where N = total number of accidents; x_i = linear coordinate of the i th accident point (unit: m); and $L_{e,i}$ = equivalent loss of the i th accident.

Dual-Dimensional Coupling and Risk Grading

The quantile method is used to divide the continuous probability intensity $P(x)$ and severity intensity $I_{\text{sev}(x)}$ into three levels: Low (L), Medium (M), and High (H). The 33.3% and 66.6% quantiles are adopted as thresholds. This method can adapt to data distribution and reduce the subjectivity of fixed thresholds. It should be clarified that the results obtained by this method represent the relative risk ranking within the study road section, not the absolute risk level—a road section classified as high risk only indicates that it ranks high in risk within the study scope, and calibration with regional benchmark values is required when extending to a larger road network.

To integrate the dual-dimensional level results and obtain a comprehensive risk indicator, a decision-oriented risk scoring formula is constructed as follows:

$$\text{Score} = 3 \times S + P + 1 \quad (8)$$

where S and P = values corresponding to the severity level and probability level, respectively (Low = 0, Medium = 1, High = 2). This formula assigns a weight of three to severity (S) compared to probability (P), reflecting the “consequence-priority” principle of safety management and focusing on identifying high-risk road sections with low frequency and high loss. This weight setting is subjective, and the impact of different weight schemes (e.g., $\text{Score} = S + P + 1$, $\text{Score} = S \times P + 1$) on risk grading will be compared later.

The scoring results (1–9 points) form a 3×3 risk matrix (Table 3), which can be further merged into five-level comprehensive risk ratings (Levels I–V) to facilitate engineering application and priority ranking (Table 4).

Table 4. Conversion guidelines for initial risk score to comprehensive risk level initial score

Range	Comprehensive risk level	Risk description	Risk response strategy
1–2	Level I	Low risk (both probability and severity are low)	Within the safety threshold, no special governance measures required
3–4	Level II	Lower risk (driven by single-dimension medium-low levels)	Implement basic prevention and control to block the evolution of risk
5–6	Level III	Medium risk (balanced dual dimensions or single-dimension high)	Allocate conventional resources to establish preventive control mechanisms
7–8	Level IV	Higher risk (driven by high consequences or high probability)	Prioritize resource allocation for targeted risk mitigation
9	Level V	Coupling of high probability and high severity, significant risk chain effects	Urgently allocate resources and launch special governance plans

Empirical Analysis and Model Validation

Characteristics of Empirical Data and Limitations of Traditional Methods

Data Source, Acquisition Process, and Accuracy Verification

To verify the engineering applicability of the proposed model, a typical Class II mountain highway in Guangxi was selected for the case study. This section spans a mileage stake range from K30+000 to K53+000, with a total length of 23 km, no median divider, a design speed of 60 km/h, and an annual average daily traffic (AADT) of approximately 8,520 passenger car units (pcu).

Accident data were collected from the official accident records of the local traffic management department, covering three complete years (2021–2023). The data were subjected to standardized and desensitization processing, following a standardized workflow, in five steps. First, retrieve all accident records of the case section; second, screen road traffic accidents within the highway right-of-way and exclude nonroad accidents; third, verify the completeness of core attributes in the records; fourth, calibrate accident location coordinates; fifth, desensitize sensitive information such as accident parties and vehicles, retaining only public attributes for analysis. After data cleaning, 46 valid accident records were obtained, each including core fields such as accident mileage stake (calibrated by both GPS and road linear reference system), accident time, number of casualties, and direct property loss in ¥10,000. Table 5 provides a summary of core statistical data. The accidents exhibit significant spatial heterogeneity in distribution, and casualty accidents show a strong correlation with sections with complex alignments.

Data Quality Control and Completeness Verification

The overall missing rate of core attributes of accident data (location, time, casualties, and direct economic loss) is relatively low, with only sporadic missing values observed in direct economic loss. Stratified mean imputation was adopted, where data were stratified by road segment division + accident type (with/without casualties) and missing values were filled with the mean value within each

Table 5. Accident statistics for a secondary road in Guangxi (K30+000–K53+000) from 2021 to 2023

Location (km)	Casualties (persons)	Loss (¥10,000)	Location (km)	Casualties (persons)	Loss (¥10,000)	Location (km)	Casualties (persons)	Loss (¥10,000)
30.71	None	0.15	37.02	None	4.5	45.35	None	1.1
32.12	Injured 2	3.2	37.08	Injured 1	6.5	45.57	Injured 1	2.3
32.33	None	0.9	37.21	None	1	45.74	Dead 1	8.5
32.57	Injured 1	2.12	37.24	None	1.2	46.03	None	0.8
33.06	None	1.5	37.55	Injured 2	5	46.21	None	0.6
33.24	Injured 1	2.8	37.57	Injured 3	7.2	47.12	None	0.7
34.52	None	1.3	38.56	Injured 1	1.8	47.32	Injured 1	1.1
34.77	Injured 2	5.61	39.2	Dead 2	12	47.65	Injured 1	2.4
35.01	None	0.8	39.52	Injured 1	1.5	47.87	None	0.5
35.29	Injured 1	2	40.55	None	3.8	48.06	None	0.6
35.57	None	1	41.13	Injured 1	1.6	48.26	Injured 1	2.1
35.83	Dead 1, Injured 1	9.5	42.83	None	1.4	48.86	None	0.3
36.01	None	1.2	43.25	Injured 1	1.92	49.07	None	0.22
36.22	Injured 1	2.5	44.02	None	0.5	49.34	None	0.4
36.57	None	0.9	44.23	Injured 1	1.32	50.05	None	0.33
36.76	Injured 2	4.83	44.54	None	0.7	52.18	None	0.15

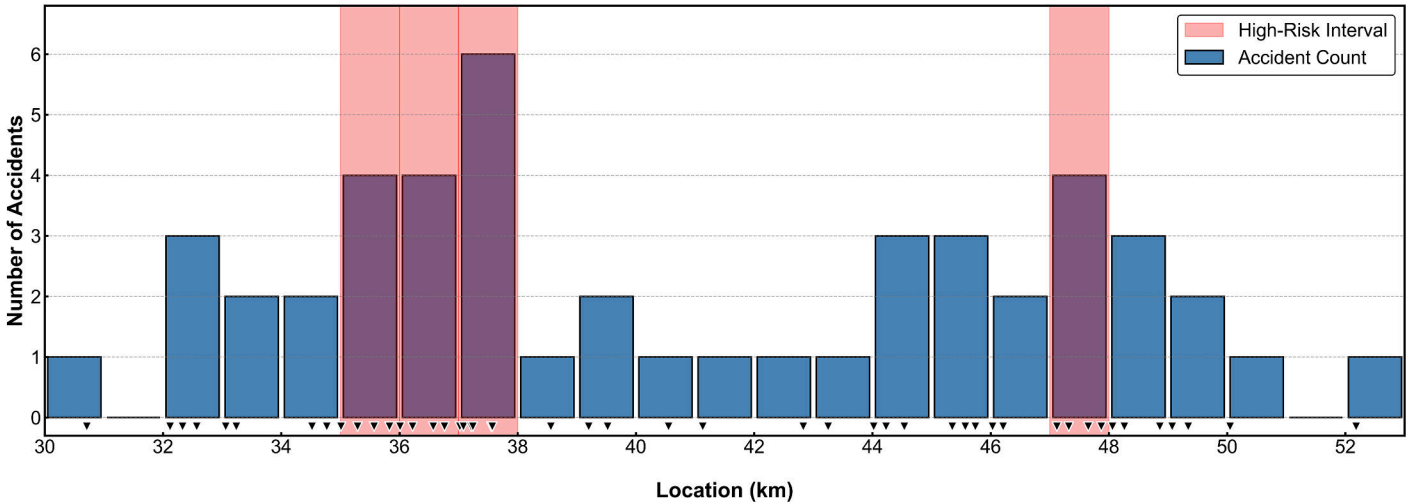


Fig. 3. Accident frequency distribution in 1-km segments (high-risk sections: four or more accidents).

stratum to avoid cross-stratum bias. Through cross-verification with traffic police records, road maintenance logs, and emergency department records of hospitals along the route, no significant underreporting was identified.

Reproduction of Limitations of Traditional Discrete Methods
To intuitively reveal the defects of traditional fixed-unit methods, Figs. 3 and 4 present accident frequency statistics with unit lengths of 1 and 0.5 km, respectively. Comparisons show the following: At the 1-km scale (Fig. 3), sections such as K35–K37 and K47 were identified as high-risk units; when refined to the 0.5-km scale (Fig. 4), the locations and scopes of risk hotspots changed significantly, with some risks further concentrated in shorter segments. This directly highlights the scale sensitivity issue of this method—differences in division scales can lead to inconsistent risk identification results. In addition, regardless of the scale, the risk within each unit is assumed to be uniformly distributed, which not only fails to reflect the risk gradient change but also completely ignores the impact of accident points crossing unit boundaries, exposing its inherent limitation of spatial representation distortion.

Data Processing Workflow and Reproducibility Statement
To ensure the transparency and reproducibility of the method, Fig. 5 illustrates the full workflow from raw data input to the final risk field generation, with specific steps as described below.
First, for data preprocessing, the raw data are cleaned to exclude nonroad accidents, and accident mileage stake coordinates are calibrated using the GPS + linear reference system to generate a standardized data set. Second, for parameter calibration, σ is determined to be 116.7 m based on road section attributes (without median divider, design speed of 60 km/h, corresponding to a passing sight distance of 350 m). Third, for risk field calculation, the continuous risk probability intensity field $P(x)$ and severity intensity field $I(x)$ are calculated respectively based on Eqs. (5)–(7). Fourth, for risk grading, $P(x)$ and $I(x)$ are discretized using the 33.3% and 66.6% quantiles to generate a dual-dimensional three-level classification; Eq. (8) and the rules in Tables 4 and 5 are combined to generate five-level comprehensive risk ratings. Fifth, for analysis and validation, parameter sensitivity analysis (testing the impact of 2σ , 3σ , and 4σ values) and comparative verification with benchmark methods such as fixed-bandwidth KDE are conducted.

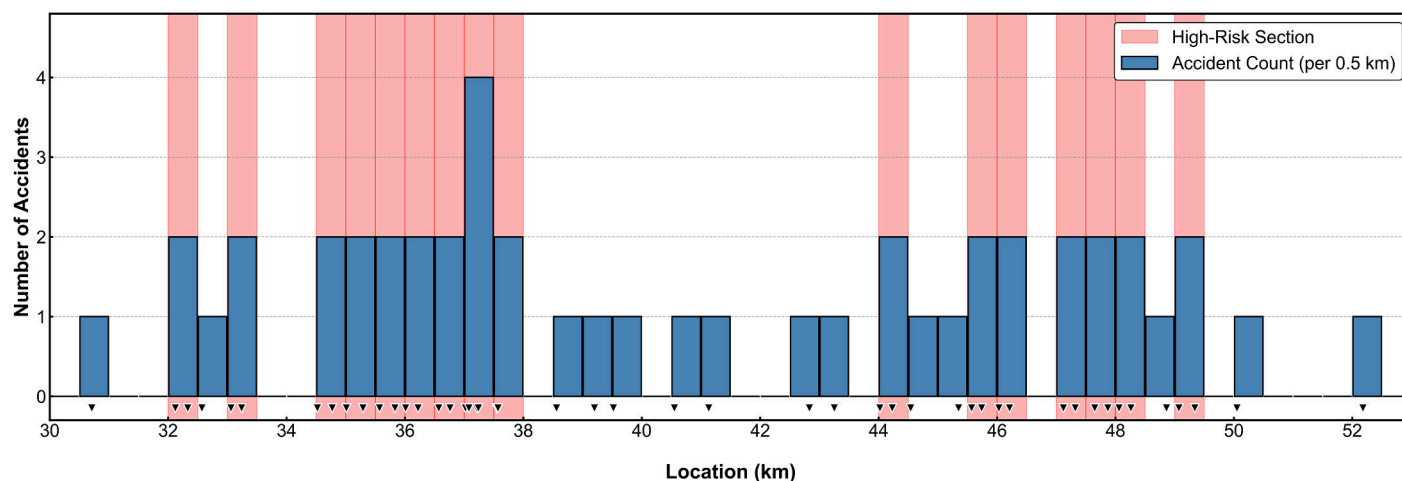


Fig. 4. Accident frequency in 0.5-km segments (high-risk sections: two or more accidents).

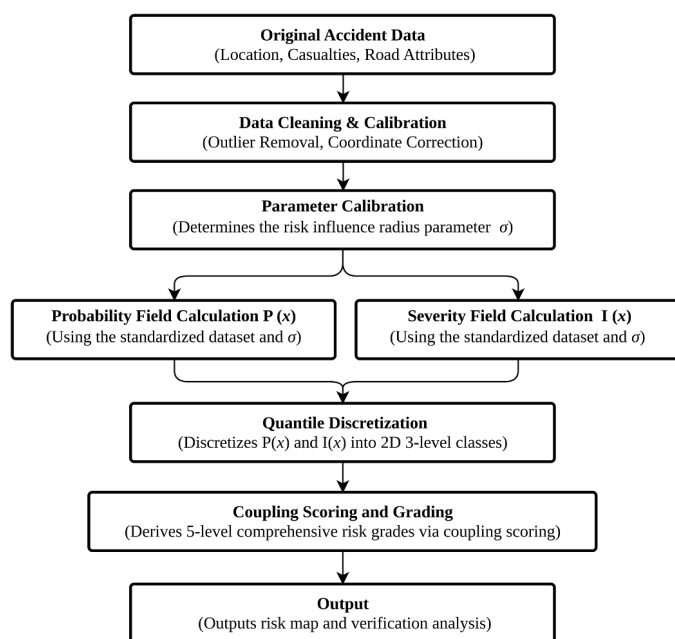


Fig. 5. Flowchart of data processing and risk field generation.

All numerical calculations, model construction, visualization, and statistical analyses in this study were completed in the Python 3.9 environment. To ensure the reproducibility of analyses and reliability of results, several open-source libraries were used in the calculation process, including NumPy 1.21.0 for array and numerical computation, SciPy 1.7.0 for Gaussian kernel estimation (via the `gaussian_kde` function), pandas 1.3.5 for data cleaning, imputation, and standardization, Matplotlib 3.4.3 and Seaborn 0.11.2 for risk field visualization and chart plotting, and scikit-learn 1.0.2 for the calculation of verification indicators such as quantiles and area under the curve (AUC). All these tools can be installed via pip, which ensures the consistency of the experimental environment and reproducibility of the analyses.

Results of Risk Field Construction

Gaussian Superposition Construction of Risk Probability Intensity Field

Based on Eq. (5), 46 accidents were treated as independent risk sources for Gaussian superposition, constructing a continuous

risk probability intensity field. Fig. 6 first presents the Gaussian curves of risk intensity attenuation with distance for three representative accident points as individual risk sources, intuitively reflecting the risk influence range defined by the standard deviation $\sigma = 116.7$ m (corresponding to an influence radius $D = 350$ m).

The superposition of Gaussian curves from all accident points forms the continuous risk probability intensity field shown in Fig. 7. Its key spatial characteristics are as follows: (1) the superposition of curve peaks in accident-dense segments forms significant risk peaks, truly reflecting the accident aggregation effect; (2) from the core peak area outward, the risk intensity attenuates smoothly and continuously (gradually changing from dark red to light red), naturally characterizing spatial risk attenuation and mitigating the abrupt boundary changes of fixed-unit methods; (3) the curves of isolated and sparse accident points form local protrusions, revealing the potential impacts of nonhotspot accidents and addressing the shortcoming of traditional methods in insufficiently focusing on scattered accidents.

Construction of Accident Severity Intensity Field

The construction of the accident severity intensity field aims to quantify the spatial distribution of accident consequences. First, the equivalent loss (L_e) of each accident was calculated based on Eq. (6). Herein, the key parameters α and β serve as casualty loss conversion coefficients, which convert the consequences of injuries and deaths into comparable economic equivalents, facilitating the dual-dimensional coupling of probability-severity. The values of these coefficients reflect social cognition of the differentiated socio-economic costs of casualty accidents. With reference to the judicial practice of personal injury compensation for road traffic accidents in the case area, this study sets α at ¥50,000 per person (for injuries) and β at ¥500,000 per person (for deaths). This assignment exhibits certain regionality and subjectivity, and the sensitivity of these values will be discussed in detail later.

Subsequently, using Eq. (7) and the same standard deviation as the probability field ($\sigma = 116.7$ m), a continuous severity intensity field was constructed through Gaussian kernel superposition. Fig. 8 shows the attenuation profiles of severity with distance for individual accidents (especially those with high L_e values).

After the superposition of all accident points, the total severity intensity field was derived, as illustrated in Fig. 9. Compared with the probability field, this severity field exhibits a spatial pattern that is both correlated and distinct. For instance, at locations where

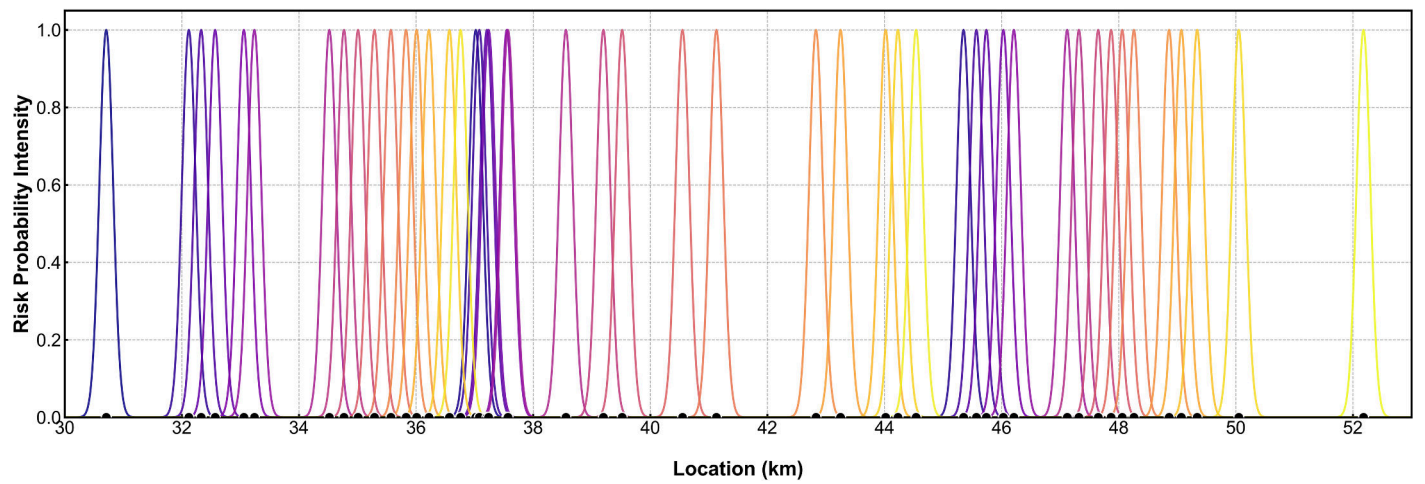


Fig. 6. Individual Gaussian profiles prior to superposition (Gaussian influence radius: 350 m).

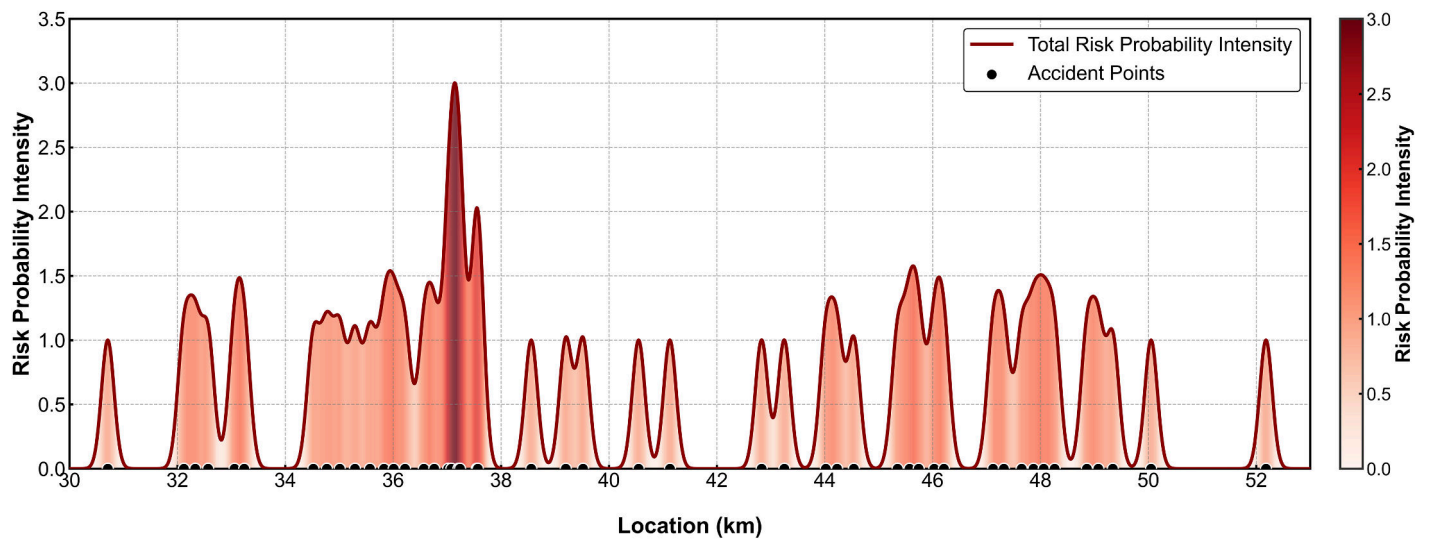


Fig. 7. Total risk intensity following Gaussian superposition (Gaussian influence radius: 350 m).

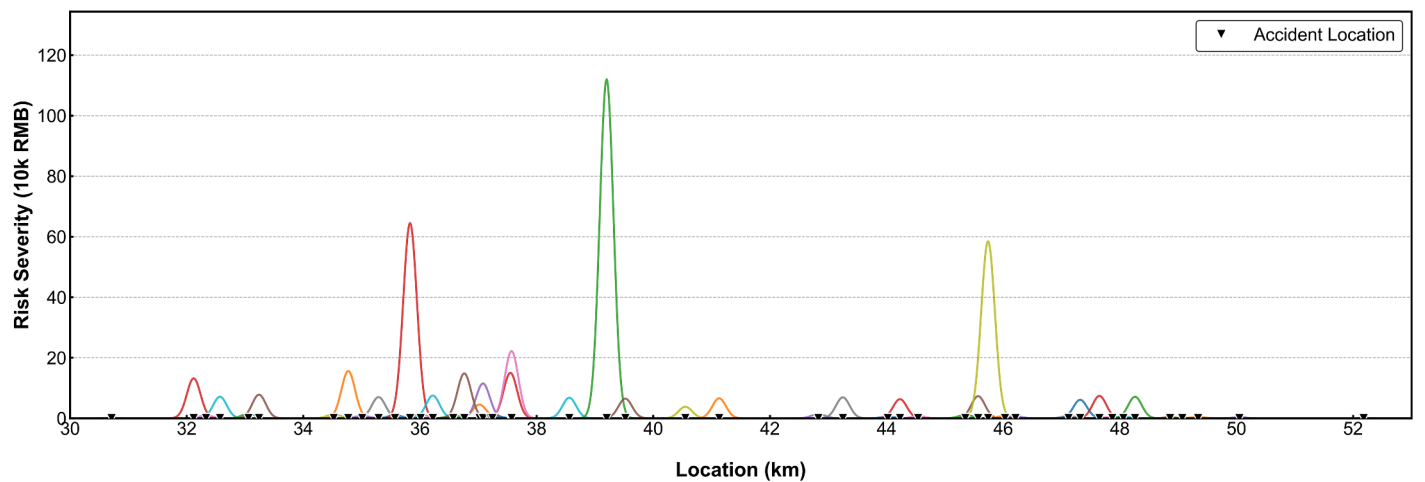


Fig. 8. Individual risk profiles prior to superposition (influence radius: 350 m).

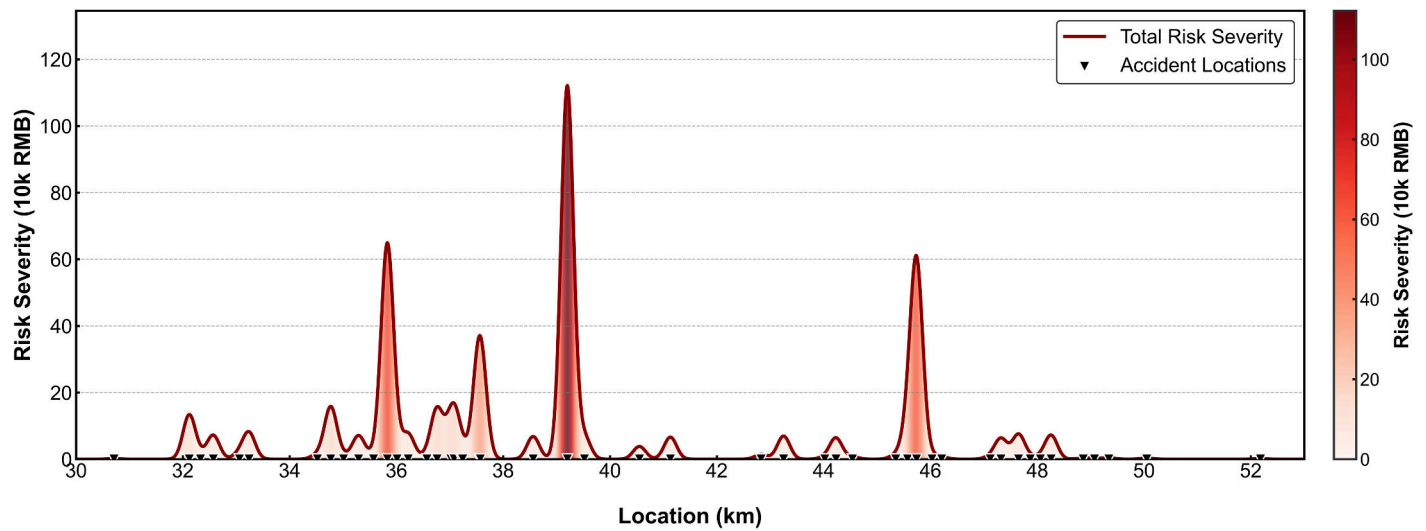


Fig. 9. Total risk intensity following Gaussian superposition (influence radius: 350 m).

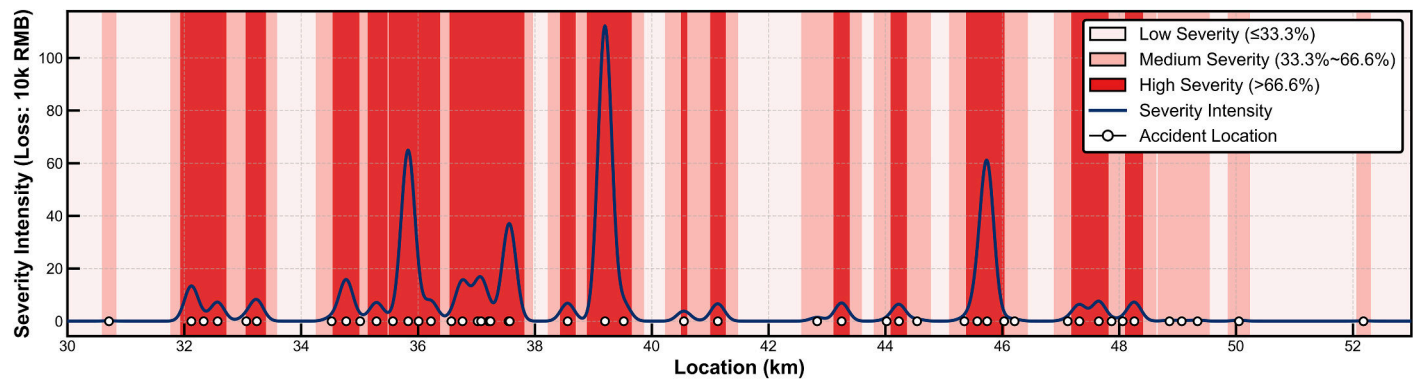


Fig. 10. Risk probability distribution using a three-tier risk heatmap.

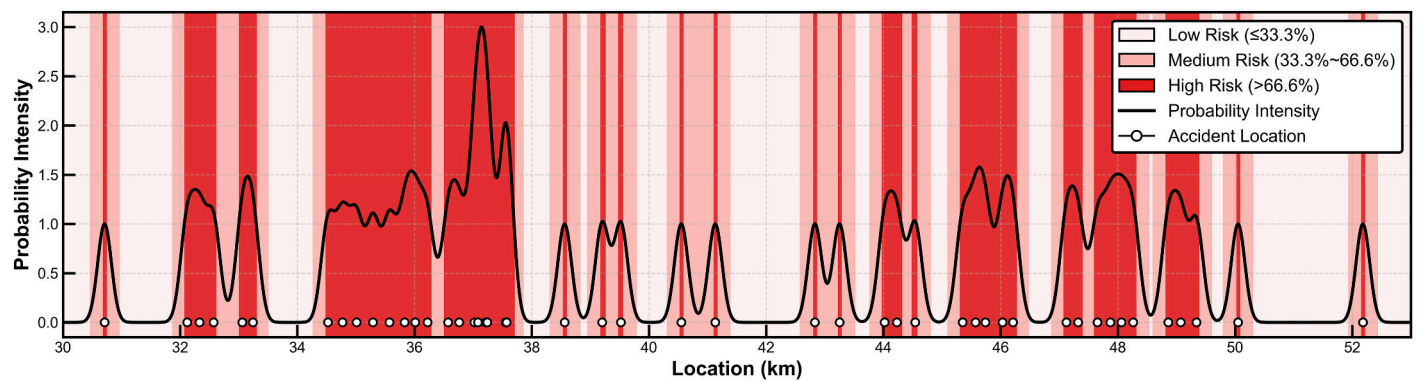


Fig. 11. Distribution of risk severity using a three-tier risk heatmap.

fatal accidents occurred, even if the accident frequency is not high, significant isolated peaks will form owing to the extremely high values of L_e , intuitively demonstrating that relying solely on accident frequency will severely underestimate the risk level of such low-frequency, high-loss road segments. In contrast, for certain road segments with frequent accident occurrences but minor consequences, the intensity of the severity field remains relatively low. This field effectively enables the transformation of the risk perspective from “accident occurrence frequency” to “accident consequence magnitude.”

Dual-Dimensional Coupling Evaluation Results

Dimension Discretization and Risk Matrix Construction

To support comprehensive risk decision making, the continuous probability field $P(x)$ and severity intensity field $I_{sev(x)}$ were first discretized into three levels [Low (L), Medium (M), High (H)] based on their 33.3% and 66.6% quantiles. The classification results are presented as heatmaps in Figs. 10 and 11, respectively. This relative grading method can adapt to the data distribution characteristics of the study section.

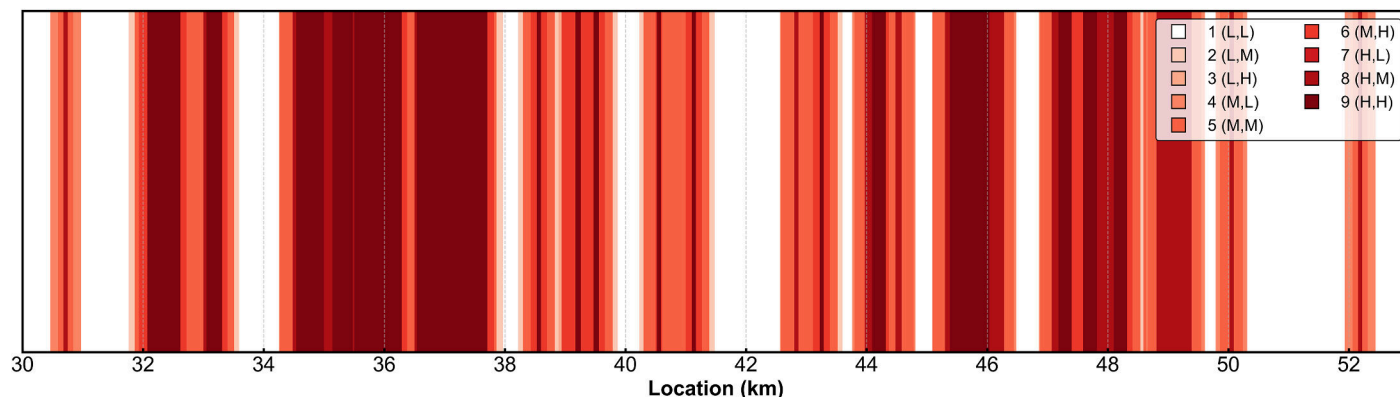


Fig. 12. 3×3 risk matrix classification (prior to consolidation into 5 levels).

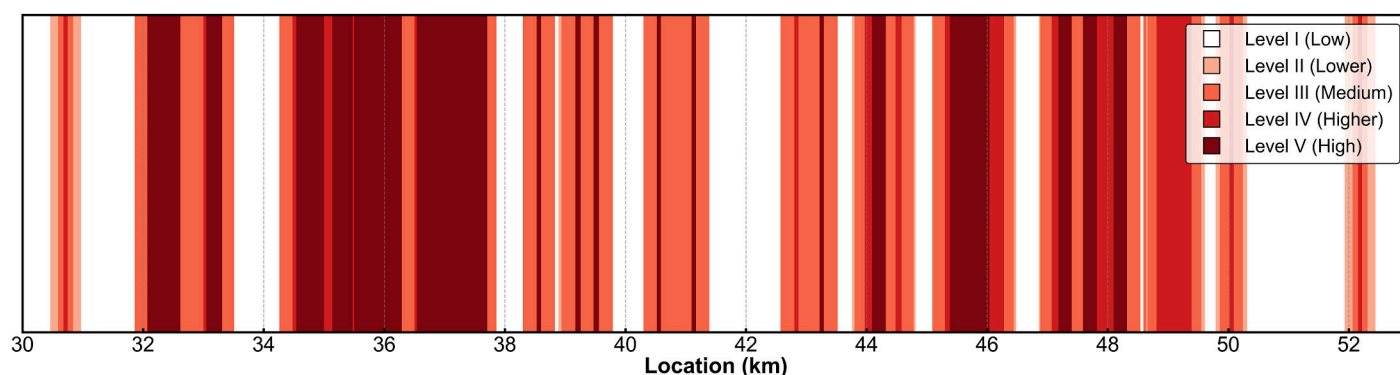


Fig. 13. Final 5-level road risk assessment (analysis range: K30+000-K53+000).

Table 6. Analysis of quantitative characteristics of risk levels

Risk description	Entire section	Risk Level I	Risk Level II	Risk Level III	Risk Level IV	Risk Level V
Evaluation indicator	23	7.60	0.92	6.79	2.10	5.59
Risk probability cumulative sum (dimensionless)	14.04	0.09	0.2	3.67	2.48	7.52
Risk probability proportion of entire section (%)	100	0.64	1.42	26.14	17.66	53.56
Risk severity cumulative sum (¥10,000 × km)	129.95	0.4	0.34	36.49	2.83	89.71
Risk severity proportion of entire section (%)	100	0.31	0.26	28.08	2.18	69.03

Note: The probability integral is a dimensionless metric derived by multiplying the accident probability density by the length of the road section; the severity integral, expressed in ¥10,000 units, is determined by multiplying the economic damage by the length of the road section.

Subsequently, according to the decision rule defined in Eq. (8) ($\text{Score} = 3 \times S + P + 1$), the probability level (P) and severity level (S) of each road point were coupled into an initial risk score ranging from 1 to 9, which was then mapped to a 3×3 risk matrix. The results are shown in Fig. 12, which clearly illustrates the spatial distribution of different level combinations.

Comprehensive Risk Rating Division

In accordance with the conversion rules in Table 4, the nine initial score categories were finally merged into five-level comprehensive risks (Level I: Low Risk; Level V: Extremely High Risk). The spatial distribution of the final risk ratings along the road is shown in Fig. 13. The mileage proportion and risk indicator contribution of each grade section are summarized in Table 6.

Based on the dual-dimensional risk assessment model, sections of different risk grades exhibit significant discrepancies. Level V sections, accounting for 24.3% of the total road length, concentrate 53.56% of the total probability risk and 69.03% of the total severity risk. They are mainly distributed in sections with continuous sharp

bends and limited sight distance, serving as key target areas for road safety management. Level IV sections, accounting for 9.1% of the total mileage, are characterized by high frequency but low loss, contributing 17.66% of the total probability risk but only 2.18% of the total severity risk. These sections are mostly located at certain intersections, and the model can avoid excessive intervention on them. Level III sections exhibit basically consistent proportions of risk and mileage, classed as medium-risk sections that require regular monitoring. Level I and II sections, combining to account for 37.04% of the total mileage, have extremely low total risk contributions and are classified as low-risk sections. These results verify that the dual-dimensional model can effectively distinguish different risk types and provide support for differentiated road safety management.

Comparative Analysis with Benchmark Methods

To verify the effectiveness of the proposed dynamic Gaussian superposition risk field model, mainstream fixed-bandwidth KDE

and fixed-segmentation statistical methods were selected as benchmarks for quantitative comparison across four dimensions: recognition accuracy, data correlation, gradient smoothness, and error (Figs. 14 and 15).

In terms of recognition accuracy (AUC), the AUC values of the probability and severity fields of this model are 0.9641 and 0.8669, respectively. This represents an increase of 12.79% and 15.24% compared with the fixed-bandwidth KDE method, and 15.31% and 5.88% compared with the fixed-segmentation method, respectively. These results indicate that the dynamic standard deviation ($\sigma = 116.7$ m) can more accurately capture risk aggregation characteristics, alleviating the “one-size-fits-all” attenuation of the fixed bandwidth and the boundary disruption effect of the fixed-segmentation method.

Regarding data correlation, the correlation coefficient of the model’s probability field (0.4772) outperforms those of the two benchmark methods; the correlation coefficient of the severity field (0.9968) is close to the optimal value and significantly higher than that of the fixed-bandwidth KDE method (an increase of 9.32%), effectively addressing the latter’s insufficiency in responding to high-consequence accidents. In terms of gradient smoothness, the smoothness index of the model’s probability field (0.0068) is 2.34 times and 1.66 times those of the fixed-bandwidth KDE method and the fixed-segmentation method, respectively. It can more accurately reflect the continuous attenuation of risk with distance, avoiding the “step effect” and local oversmoothing.

Error analysis shows that with the model’s results as the benchmark, the root mean square errors (RMSEs) of the fixed-bandwidth KDE method and the fixed-segmentation method are 0.1637 and 0.1433, respectively, indicating that the model has better output stability. In summary, by adapting the dynamic standard deviation to the actual road characteristics (e.g., $D = 350$ m for roads without median dividers), the model maintains the spatial continuity of the risk field while exhibiting good engineering applicability. It can

provide a reliable foundation for accurate identification of high-risk road sections and optimal allocation of safety resources.

Sensitivity Analysis

To verify the robustness and management applicability of the proposed risk field model under key parameters and settings, a systematic multidimensional sensitivity analysis was conducted, covering core spatial parameters, loss quantification coefficients, risk assessment weights, and stability under small-sample conditions.

First, regarding parameter sensitivity, the model exhibits distinct sensitivity characteristics to key parameters. For the spatial standard deviation σ , as the core parameter controlling the risk influence range and smoothness, its values—with a test range corresponding to $D/4$ to $D/2$ —mainly adjust the spatial gradient of the risk field. Larger σ values result in smoother risk distributions with wider influence ranges, while smaller σ values concentrate risks more in areas adjacent to accident points. Notably, the spatial locations of the highest risk level sections (Levels IV and V) identified under different σ schemes are highly consistent, confirming the robustness of the model’s core risk identification results. This parameter can serve as an engineering interface to adjust the system’s “risk alertness,” calibrated according to management tolerance for risk spread scope. For the loss conversion coefficients α and β , the impact of different monetization coefficients for casualties was tested with α ranging from ¥30,000 to ¥70,000 per person and β ranging from ¥300,000 to ¥700,000 yuan per person. Analysis shows that when the value ratio of death to injury losses (β/α) remains constant, proportional scaling of the coefficients has a minimal impact on risk grading results based on quantile division, reflecting the model’s insensitivity to absolute economic loss valuation. However, adjusting the relative ratio—such as significantly increasing β —directly enhances the prominence of sections with fatal accidents in the severity field and final comprehensive risk,

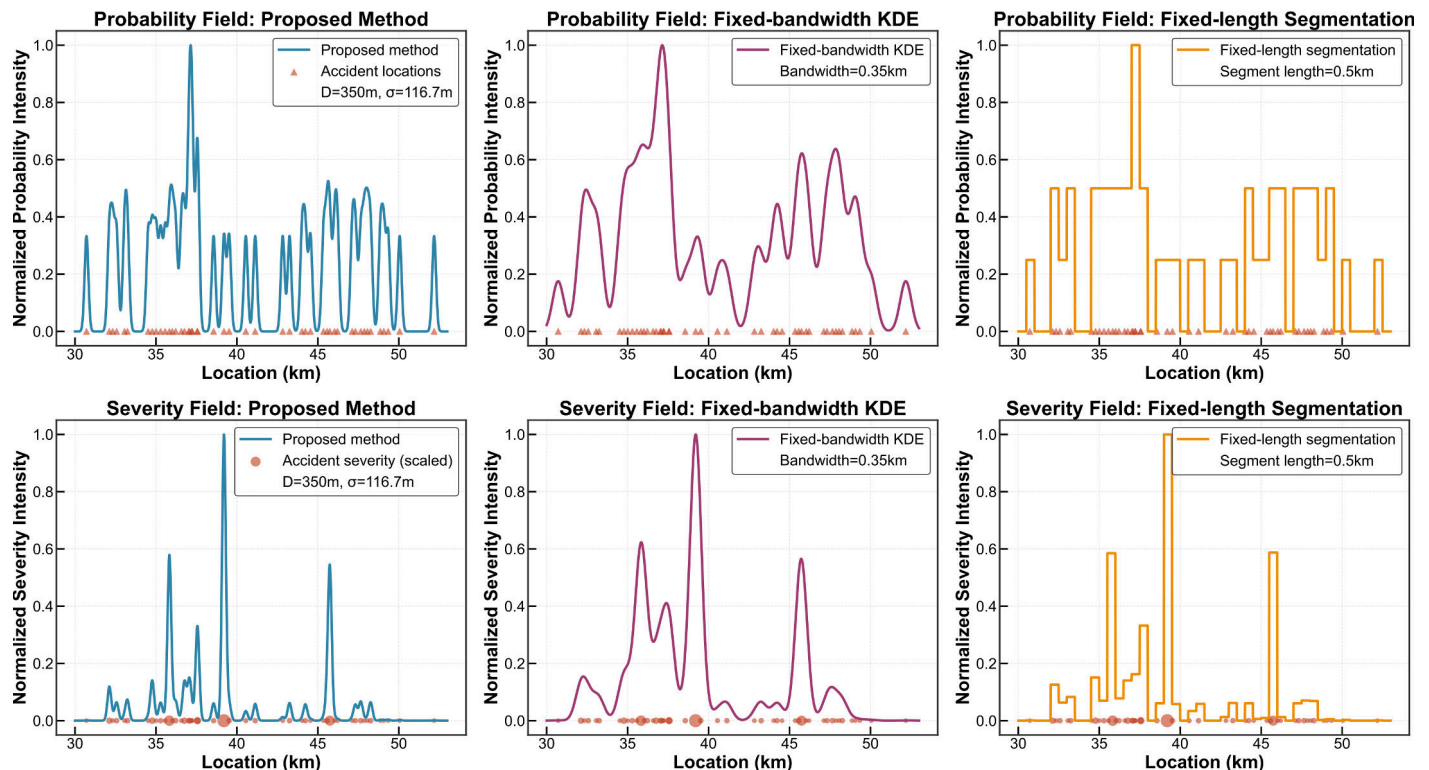


Fig. 14. Comparative analysis of road risk assessment methods.

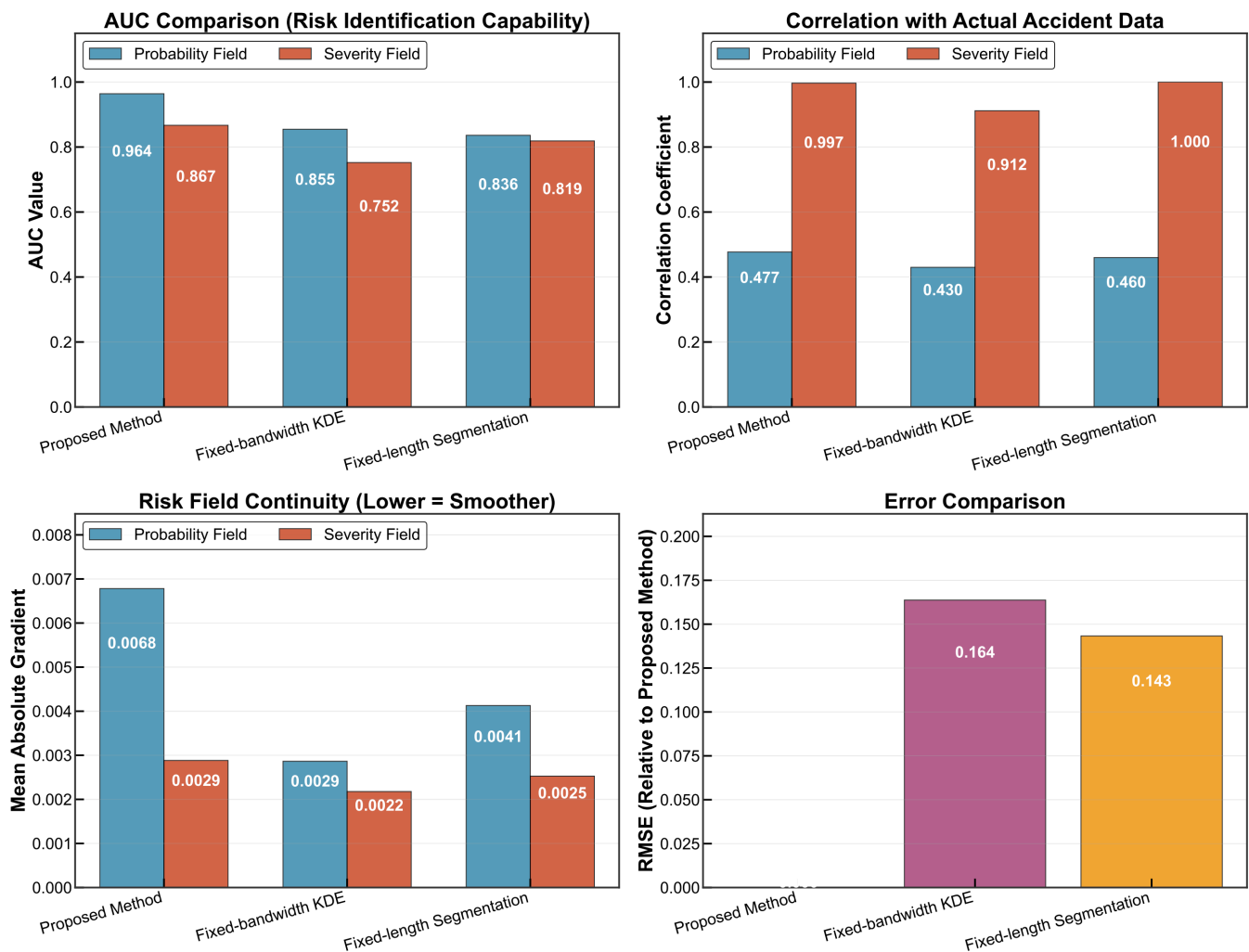


Fig. 15. Comprehensive performance comparison of risk assessment methods.

which aligns with the safety management principle of prioritizing life loss. This also indicates that β is a key subjective parameter reflecting management value orientation.

Second, for the comparison of risk scoring weight schemes, the coupling method of probability (P) and severity (S) in risk assessment directly affects the risk grading orientation. Three weight schemes were compared, including the original scheme (Score = $3S + P + 1$, consequence-prioritized), the equal-weight scheme (Score = $S + P + 1$), and the product scheme (Score = $S \times P + 1$). Validation based on road sections with actual fatal/severe injury accidents shows that the original scheme achieves the highest recognition rate for these sections. It can most effectively highlight high-consequence sections, aligning with the needs of emergency management for preventing major accidents. The equal-weight scheme results in a more balanced risk distribution, suitable for scenarios with balanced resource allocation. The product scheme is extremely sensitive to high-frequency high-loss sections, but may weaken the identification of low-frequency high-loss risk points. The choice of weight should match management objectives and resource constraints.

Third, regarding the discussion on stability under small-sample conditions, this empirical case is based on data from 46 accidents over three years on a 23-km Class II highway, representing a typical small-sample scenario. Despite the limited sample size, the accidents on the section exhibit sufficient

spatial density (approximately 0.67 accidents per km per year) and aggregation characteristics, providing a basis for method verification. The aforementioned sensitivity analysis confirms that the core high-risk areas identified by the model are robust to parameter changes. It should be noted that this study focuses on proposing and validating a methodological framework integrating spatial attenuation and multidimensional coupling through a case study. The quantile grading method adopted in this case achieves effective relative risk differentiation, but its thresholds are data dependent. The generalizability of the model and more accurate threshold calibration require further verification and calibration with broader and more diverse road network data.

Conclusions and Future Work

Conclusions

To address the issues of missing spatial continuity of risk, boundary distortion, and information “averaging” in traditional fixed-unit statistics for identifying accident-prone road sections, this study proposes a road accident risk field reconstruction model integrating dynamic spatial attenuation and probability-severity dual-dimensional coupling. Through theoretical construction and empirical analysis on a Class II mountain highway in Guangxi, the core conclusions are summarized below.

First, a dynamic risk probability field transformation framework with clear engineering significance is constructed. Based on the principle of information diffusion and Gaussian KDE, a dynamic standard deviation function is proposed, which establishes a potential physical correlation between the Gaussian kernel standard deviation and road design speed, cross-sectional type, as well as drivers' sight distance requirements. This shift helps to change the determination of key model parameters from pure mathematical optimization to engineering-oriented calibration, which may enhance the interpretability and engineering applicability of the risk field to a certain extent and assist in addressing the drawback of "prioritizing fitting over mechanism" in traditional continuous risk mapping.

Second, a continuous risk assessment system coupling probability and severity dual dimensions is established. By integrating casualties and direct economic losses through an accident equivalent loss function, a continuous accident severity field is constructed. Combined with the quantile method and risk matrix, the two fields are discretized and coupled to generate comprehensive risk ratings (Levels I–V), which can help to accurately distinguish multiple risk patterns such as high-frequency low-loss, low-frequency high-loss, and high-frequency high-loss. This provides a potential quantitative basis for differentiated safety management and may help to alleviate the deficiencies of traditional multidimensional evaluation methods, such as crude coupling logic and neglect of the "consequence-priority" demand.

Empirical verification indicates that the model exhibits certain advantages: the boundaries of the risk field fit the road alignment relatively well, with a probability field gradient smoothness of 0.0068, which helps to mitigate the boundary and step effects of the fixed-unit method; compared with fixed-bandwidth KDE and fixed-length segmentation method, the model achieves a probability field AUC of 0.9641, a severity field AUC of 0.8669, and a Pearson correlation coefficient of 0.4772 between the probability field and accident data, suggesting notable improvements in identification accuracy and data fit. In addition, the model shows good adaptability in the small-sample scenario with only 46 accidents, as it can supplement spatial risk information through information diffusion and match onsite hazard factors to a certain degree.

In summary, this model contributes to the evolution of road risk assessment from discrete statistics to continuous field analysis, provides some improvements to the theories and methods of continuous risk mapping and multidimensional safety evaluation, and may offer reliable technical support for highway safety hazard investigation, optimal allocation of safety resources, and differentiated safety management.

Future Work

Considering the limitations of this study, future work will focus on model optimization and expansion, attempting to further enhance its generalizability and practicality in a gradual manner:

Future work can verify and tentatively optimize the generalizability of core assumptions. Collect empirical data from diverse scenarios with different accident types and risk sources to verify the rationality of the assumption (i.e., the risk impact radius is equivalent to the design sight distance), and revise relevant parameters accordingly if necessary.

Future studies can attempt to expand the model's generalization ability. Plan to conduct multicase studies on plain highways, expressways, urban roads, and regions with significantly different traffic flow characteristics. Relevant parameters will be calibrated to seek relatively stable application of the model across multiple scenarios.

Future studies can attempt to integrate multisource dynamic data to optimize the model. Consider introducing factors such as real-time traffic flow and meteorological conditions, integrate historical and real-time monitoring data, and attempt to develop a spatiotemporal dynamic risk prediction model, aiming to upgrade the model from static assessment to dynamic prediction.

Future studies should seek to improve the risk rating and cross-regional comparison system. Explore possible integration paths of machine learning and this model to better handle complex nonlinear risk relationships; based on large-scale road network data, explore the calibration method of absolute risk thresholds, seeking to support relatively unified risk assessment and management across regions and road types.

Engineering Application

The proposed dynamic risk field reconstruction and evaluation model seeks to address the limitations of traditional discrete statistics through continuous field analysis, exhibiting certain engineering practicality to a certain extent and aiming to provide multidimensional quantitative support for road safety management.

For safety hazard investigation, relying on the continuous risk field dynamically adapted to road characteristics, the model may help intuitively identify high-risk core areas and potential risk zones, complementing traditional empirical investigation and potentially helping to improve the accuracy and efficiency of identification to some degree.

For the optimal allocation of safety resources, based on the I–V level risk rating results, the model may assist in prioritizing resource investment in Level V sections with dual-high or low-frequency high-loss risks, helping to avoid excessive investment in Level IV high-frequency low-loss sections and seeking to enhance the efficiency of safety investment as much as possible.

For supporting refined maintenance decisions, by attempting to dynamically link risk ratings with maintenance strategies, the model may help implement special inspections and preventive maintenance for high-risk sections, as well as routine monitoring for medium and low-risk sections, which may further help promote the transformation of maintenance models from time-based to demand-based.

The model is built based on open-source tools with relatively standardized processes and exhibits certain portability, which may help support its potential promotion and application by traffic management and maintenance departments.

Data Availability Statement

Some or all data, models, or code that support the findings of this study are available from the corresponding author upon reasonable request.

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Author Contributions

Jianfeng Liu: Conceptualization; Methodology; Writing – original draft; Writing – review and editing. Fuyuan Luo: Investigation; Resources; Visualization. Jianqiu Chen: Project administration. Xin Peng: Resources; Visualization. Yan Chen: Resources. Xiang Zhou: Investigation. Chengxi Yu: Formal analysis; Validation.

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